How Long Should an Exam Be?

A Binomial Risk–Reward Problem

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Abstract

A short test makes a perfect score more likely, but also makes an unexpectedly poor score more likely. A long test concentrates the observed score near the student's underlying accuracy. We model this trade-off with independent Bernoulli questions, establish a convex-order comparison between test lengths, and pose two precise optimisation problems. In an illustrative example, a seven-question test maximises the probability of a perfect score while keeping the probability of a mark below 70% under 1%. The exact finite-sample problem exhibits a striking sawtooth effect that large-deviation approximations smooth away.

1 The question

A student preparing for an exam notices that getting one question wrong matters much more when an exam contains only a few questions. But a short exam also offers the best chance of getting *every* question right. Is there an optimal number of questions?

The answer depends on what "optimal" means. If the only objective is expected percentage score, every test length is equivalent in the simple model below. A meaningful finite optimum appears when a student values a perfect score but wants to limit the probability of a bad outcome.

2 A binomial model

Assume each question is answered incorrectly with probability $\varepsilon \in (0,1)$, independently of other questions, with the same probability on each question. For a test with $n \geq 1$ questions, write

$$K_n \sim \mathrm{Bin}(n, \varepsilon), \qquad S_n = 1 - \frac{K_n}{n}, \tag{1}$$

where K_n is the number of mistakes and S_n is the proportion correct. Then

$$\mathbb{E}S_n = 1 - \varepsilon, \qquad \mathrm{Var}(S_n) = \frac{\varepsilon(1-\varepsilon)}{n}. \tag{2}$$

In particular, the expected percentage does not change with n, but the distribution becomes more concentrated. The two extreme outcomes have probabilities

$$\mathbb{P}(S_n = 1) = (1-\varepsilon)^n, \qquad \mathbb{P}(S_n = 0) = \varepsilon^n. \tag{3}$$

Both become less likely as n increases. The second is usually very small for a reasonably competent student; an ordinary "bad mark" is a more useful risk measure.

For a threshold $r \in (0,1)$, define a bad result as a score *strictly below* r. Because marks lie on a lattice, this means at least

$$m(n,r) = \lfloor n(1-r) \rfloor + 1 \tag{4}$$

errors. Its exact probability is

$$B_n(r,\varepsilon) := \mathbb{P}(S_n < r) = \sum_{k=m(n,r)}^{n} \binom{n}{k} \varepsilon^k (1-\varepsilon)^{n-k}. \tag{5}$$

The strict inequality matters at scores exactly equal to the threshold.

3 Why shorter exams are intrinsically riskier

There is a stronger statement than the variance formula.

Proposition 1 (Convex-order comparison)**.** *For any convex function $\phi : [0,1] \to \mathbb{R}$ and integers $1 \le n < m$,*

$$\mathbb{E}\,\phi(S_n) \;\ge\; \mathbb{E}\,\phi(S_m). \tag{6}$$

Equivalently, the distribution of the shorter test score is a mean-preserving spread of that of the longer test.

Proof. It suffices to compare n and $n+1$. Take $n+1$ independent zero-one indicators of correct answers, and delete one of them uniformly at random. Let A be the mean of the n remaining indicators and B the mean of the original $n+1$. Conditional on all the indicators, $\mathbb{E}[A \mid \text{indicators}] = B$. Jensen's inequality gives $\phi(B) \le \mathbb{E}[\phi(A) \mid \text{indicators}]$. Taking expectations proves the assertion, since A and B have the distributions of S_n and S_{n+1}. Iterate to obtain the general case. □

Thus a convex utility of the percentage score favours a shorter test; a concave utility favours a longer test; and a linear utility is indifferent. Real grading preferences often have thresholds and are neither globally convex nor globally concave.

4 Two ways to define an optimal length

4.1 Maximise perfection subject to a risk limit

Choose a bad-mark threshold r and a maximum acceptable risk δ. The student's problem is

$$\boxed{\max_{n \in \{1,2,\dots\}} (1-\varepsilon)^n \quad \text{subject to} \quad B_n(r,\varepsilon) \le \delta.} \tag{7}$$

As the objective is strictly decreasing in n, the optimal length is simply the *smallest feasible* integer n. This does not imply that the constraint itself becomes monotonically easier with n: its feasible set can have gaps.

If $\varepsilon < 1 - r$, the law of large numbers gives $B_n(r,\varepsilon) \to 0$, so at least one feasible length exists for every $\delta > 0$. If the underlying error rate is too high for the proposed mark threshold, there need not be any feasible length at all.

4.2 Reward a perfect score and penalise a bad one

An alternative is an expected utility

$$U_\lambda(n) = (1-\varepsilon)^n - \lambda B_n(r,\varepsilon), \qquad \lambda > 0. \tag{8}$$

The coefficient λ says how costly a bad score is compared with the benefit assigned to a perfect one. This objective can have a finite interior optimum even though there is no explicit cost per question. Unlike the constrained problem, however, the answer depends on a subjective parameter λ.

5 Worked example: a seven-question exam

Suppose $\varepsilon = 0.05$, a bad mark is *below* 70% ($r = 0.70$), and the student allows at most a 1% chance of a bad mark ($\delta = 0.01$). Direct evaluation of (3) and (5) gives:

Questions n	$\mathbb{P}(100\%)$	$\mathbb{P}(S_n < 70\%)$
1	95.00%	5.0000%
2	90.25%	9.7500%
3	85.74%	14.2625%
4	81.45%	1.4019%
5	77.38%	2.2593%
6	73.51%	3.2774%
7	69.83%	0.3757%
8	66.34%	0.5788%
9	63.02%	0.8361%
10	59.87%	0.1028%
20	35.85%	0.0034%

Every $n \le 6$ violates the 1% risk limit, while $n = 7$ meets it. Therefore $n = 7$ **is the global optimum** of (7). Its chance of a perfect score is about 69.8% and the chance of a bad score is about 0.376%.

The risk changes discontinuously when another error becomes tolerable. At $n = 6$, two errors give $4/6 \approx 66.7\%$, below 70%. At $n = 7$, two errors give $5/7 \approx 71.4\%$, above the threshold; now three errors are needed for a bad result. Hence the dramatic drop between six and seven questions, followed by another rise.

The same parameters with $\lambda = 20$ in (8) also yield $n = 7$. To establish this globally, note that $U_{20}(7) \approx 0.62320$, while for all $n \ge 10$, $U_{20}(n) \le 0.95^n \le 0.95^{10} \approx 0.59874$; direct checking of $n = 1, \dots, 9$ completes the proof.

6 Concentration and the method of types

The exact probability of an empirical error frequency $x = k/n$ is

$$\mathbb{P}(K_n = nx) = \binom{n}{nx} \varepsilon^{nx} (1-\varepsilon)^{n(1-x)}. \tag{9}$$

For types x away from the endpoints, its exponential scale is controlled by the binary relative entropy

$$D(x\|\varepsilon) = x \log \frac{x}{\varepsilon} + (1-x) \log \frac{1-x}{1-\varepsilon}, \qquad \mathbb{P}(K_n = nx) = \exp\{-nD(x\|\varepsilon) + O(\log n)\}. \tag{10}$$

Set $q = 1 - r$. When $q > \varepsilon$, a standard Chernoff bound gives

$$B_n(r, \varepsilon) = \mathbb{P}(K_n/n > q) \leq \exp\{-nD(q\|\varepsilon)\}. \tag{11}$$

In contrast, the exact probability of perfection is

$$\mathbb{P}(S_n = 1) = \exp\{-nI_0\}, \qquad I_0 = -\log(1 - \varepsilon). \tag{12}$$

The two outcomes thus have different exponential rates. Inequality (11) supplies a sufficient (not necessary) risk guarantee:

$$n \geq \frac{\log(1/\delta)}{D(q\|\varepsilon)}. \tag{13}$$

For the worked example, this bound requires $n \geq 15$, while the exact optimum is only 7. A coarse asymptotic approximation misses the integer cutoff effects that dominate short exams.

7 Questions worth exploring

1. **The sawtooth structure.** Characterise the local minima of $B_n(r, \varepsilon)$ as the cutoff $m(n, r)$ jumps. How does this depend on rational or irrational r?

2. **Utility.** For which classes of score-based utility functions is the best length finite? Convex-order comparison settles globally convex and concave utilities, but threshold utilities can behave differently.

3. **Limits.** How well do exact binomial probabilities, Chernoff bounds, and normal approximations agree at the optimum?

4. **Real exams.** Questions differ in difficulty, answers are correlated, and fatigue may change error probability with length. How much of the simple model survives these effects?

Interactive companion

An exact binomial calculator, with adjustable error probability, grade threshold, and risk tolerance, is available at

`https://grantaj.github.io/optimal-exam/`.

The associated source and the paper are at `https://github.com/grantaj/optimal-exam`.